LAB 04 / CLOUD INFRASTRUCTURE

AWS Foundations: IAM, VPC, and EC2

Identity permissions, network segmentation, and compute lifecycle

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Executive Summary

Completed introductory AWS exercises covering user and group permissions, a custom VPC, and EC2 web-server deployment. The report documents access-control checks, public and private subnet routing, instance monitoring, and protection settings.

> **SKILLS DEMONSTRATED**
>
> AWS IAM · VPC · EC2 · Security groups · CloudWatch · Apache

- Compared read-only and instance-management permissions through IAM groups.
- Configured public and private subnets, route tables, and gateways.
- Documented EC2 deployment, monitoring, resizing, and protection settings.

Evidence Included

IAM access-denial screenshots, VPC and route-table views, web-server output, and EC2 monitoring and lifecycle screenshots.

Objective

The objective was to develop practical familiarity with the AWS Management Console through exercises in identity and access management, virtual networking, and compute deployment.

Technical Background

AWS Identity and Access Management (IAM) controls access to AWS resources. Users can be assigned permissions through groups and policies; roles provide temporary access for defined tasks. The lab compares S3 read access, EC2 read access, and a limited EC2 instance-management policy.

Amazon Virtual Private Cloud (VPC) provides a logically isolated network for AWS resources. Subnets, route tables, internet gateways, NAT gateways, security groups, and network ACLs determine network placement and connectivity. Public and private subnet routing is a central focus of the exercise.

Amazon Elastic Compute Cloud (EC2) provides configurable virtual servers. The exercises use EC2 to host a web server and examine instance monitoring, network access, storage changes, instance types, and protection settings.

Procedure and Evidence

Lab 1: Introduction to AWS IAM

Task 1: Explore the Users and Groups

In the AWS search bar, search for IAM and open the IAM console.

In the left navigation pane, select IAM Users. There should be 3 users similar to below:

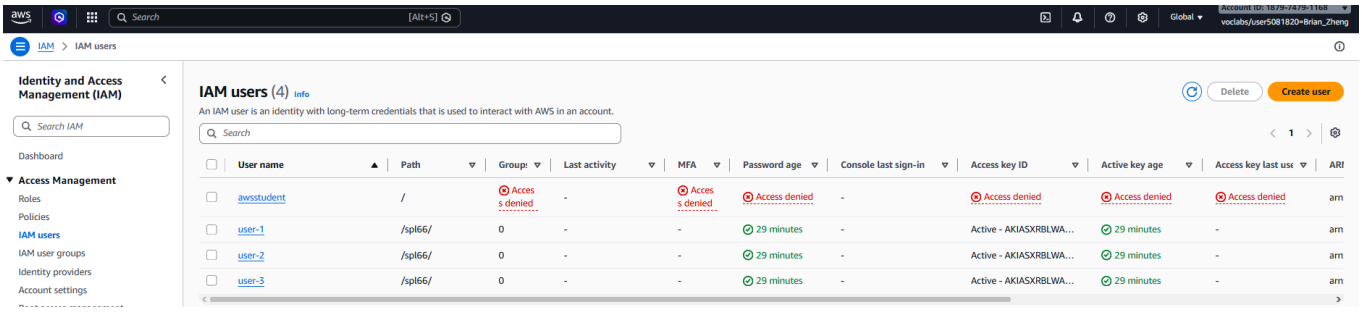


Figure 01. In the left navigation pane, select IAM Users. There should be 3 users similar to below

Select user-1. On the summary page you should see:

Permissions tab: No permissions assigned.

Groups tab: Not a member of any groups.

Security credentials tab: Has a console password.

In the left navigation pane, select User groups. The following groups should be available: EC2-Admin, EC2-Support, S3-Support

Select EC2-Support and open the Permissions tab. Here are some things you can check out:

The group has the managed policy AmazonEC2ReadOnlyAccess attached.

This policy allows users to view EC2, Elastic Load Balancing, CloudWatch, and Auto Scaling resources, but you can’t modify them.

IAM policies contain:

Effect Allow or Deny

Action Permitted AWS API operations

Resource Resources the policy applies to

Return to User groups, select S3-Support, and open the Permissions tab.

The group uses the AmazonS3ReadOnlyAccess managed policy.

Expand the policy to view details.

This policy allows users to list and retrieve Amazon S3 resources.

Collapse the policy when finished.

Return to User groups, select EC2-Admin, and click on the Permissions tab.

This group uses an inline policy, which is attached only to this specific group.

Expand the policy to view details.

The policy allows users to describe EC2 resources and start or stop EC2 instances.

Collapse the policy when finished.

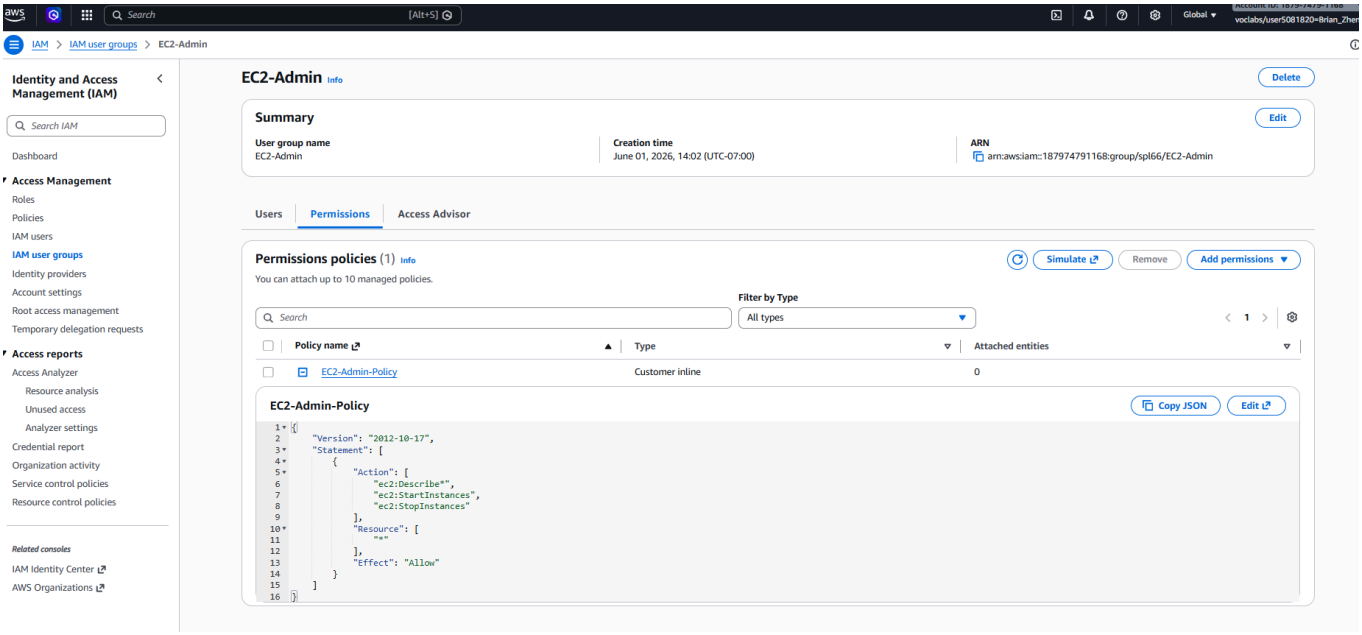


Figure 02. Collapse the policy when finished.

Task 2: Add Users to Groups

In the left navigation pane, select User groups.

Choose the S3-Support group and open the Users tab.

Select Add users, choose user-1, and then select Add users to add the user to the group.

Verify that user-1 now appears in the group's user list.

Return to User groups, select EC2-Support, and open the Users tab.

Select Add users, choose user-2, and then select Add users.

Verify that user-2 has been added to the EC2-Support group.

Return to User groups, select EC2-Admin, and open the Users tab.

Select Add users, choose user-3, and then select Add users.

Verify that user-3 has been added to the EC2-Admin group.

Return to User groups and confirm that each group displays 1 in the Users column, it should look like the following:

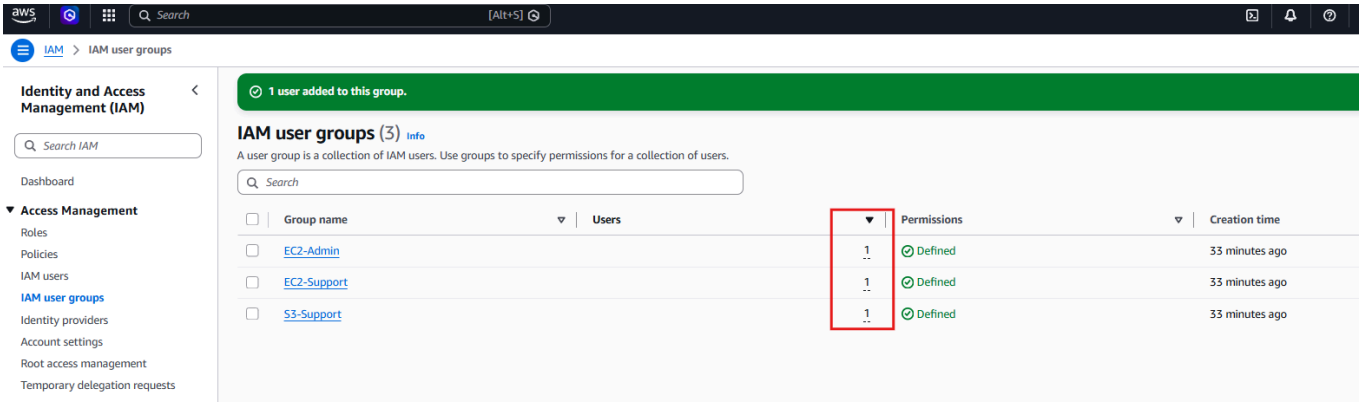


Figure 03. Return to User groups and confirm that each group displays 1 in the Users column, it should look like the following

If any group does not show 1 user, review the previous steps and ensure the following assignments are correct:

user-1 → S3-Support

user-2 → EC2-Support

user-3 → EC2-Admin

Task 3: Sign-In and Test Users

In the left navigation pane, select Dashboard.

Copy the Sign-in URL for IAM users in this account and save it in a text editor for later use.

Open a private browsing window:

Chrome: New Incognito Window

Edge: New InPrivate Window

Firefox: New Private Window

Internet Explorer: InPrivate Browsing

Test Access as user-1 (S3 Support)

Paste the IAM sign-in URL into the private browser and sign in using:

IAM user name: user-1

Password: Lab-Password1

Search for and open Amazon S3.

Open the existing S3 bucket and review its contents.

user-1 can view S3 buckets and objects because of the S3-Support permissions.

The bucket does not contain any objects.

Search for and open Amazon EC2.

Select Instances from the navigation pane.

You will receive a message stating that you are not authorized to perform this operation:

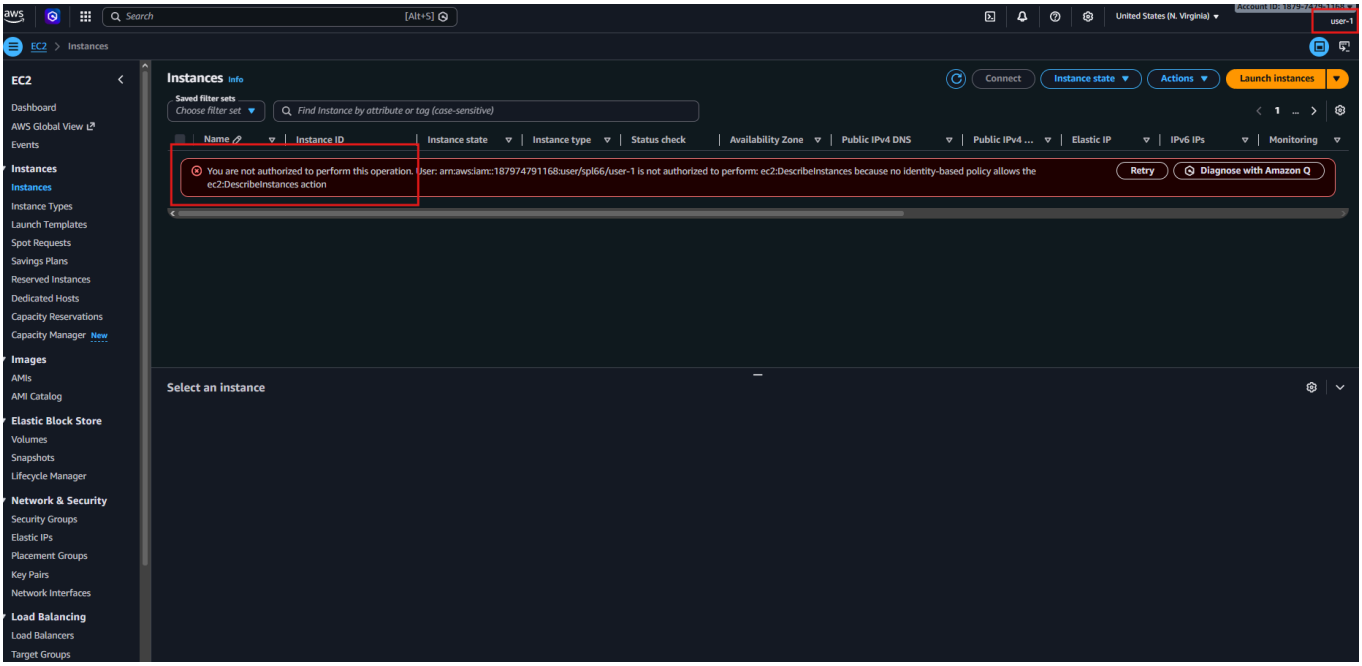


Figure 04. You will receive a message stating that you are not authorized to perform this operation

This confirms that user-1 does not have EC2 permissions.

Test Access as user-2 (EC2 Support)

Sign out of the AWS Management Console.

Return to the IAM sign-in URL and sign in using:

IAM user name: user-2

Password: Lab-Password2

Search for and open Amazon EC2.

Select Instances.

user-2 can view EC2 instances because of the EC2-Support permissions.

If no instance appears, verify that you are in the correct AWS Region.

Select the LabHost instance.

From the Instance state menu, choose Stop instance, then select Stop.

An authorization error will appear.

This confirms that user-2 has read-only access and cannot make changes to EC2 resources.

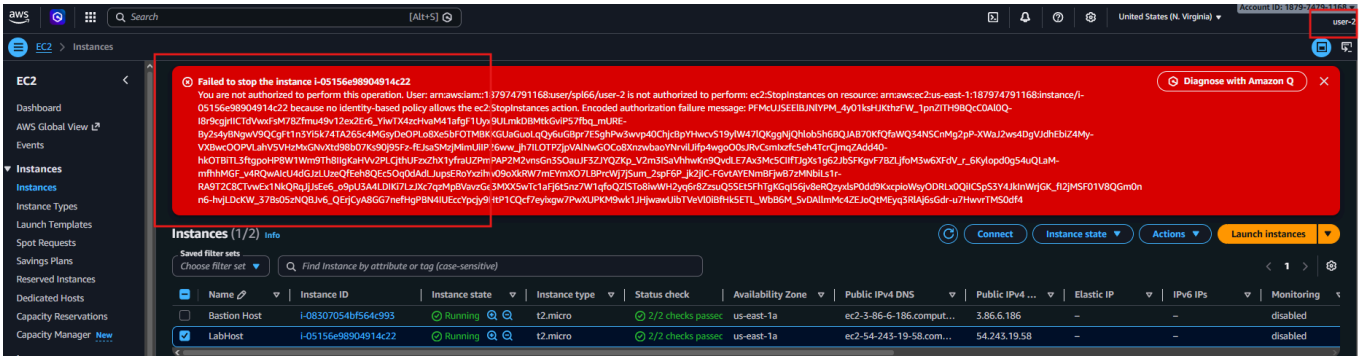


Figure 05. This confirms that user-2 has read-only access and cannot make changes to EC2 resources.

Close the error message.

Search for and open Amazon S3.

You will receive a message indicating that you do not have permission to list buckets.

This confirms that user-2 does not have S3 access.

Test Access as user-3 (EC2 Administrator)

Sign out of the AWS Management Console.

Return to the IAM sign-in URL and sign in using:

IAM user name: user-3

Password: Lab-Password3

Search for and open Amazon EC2.

Select Instances and locate the LabHost instance.

If the instance is not visible, verify that you are using the correct AWS Region.

Select the LabHost instance.

From the Instance state menu, choose Stop instance, then select Stop.

The instance will successfully enter the Stopping state like below:

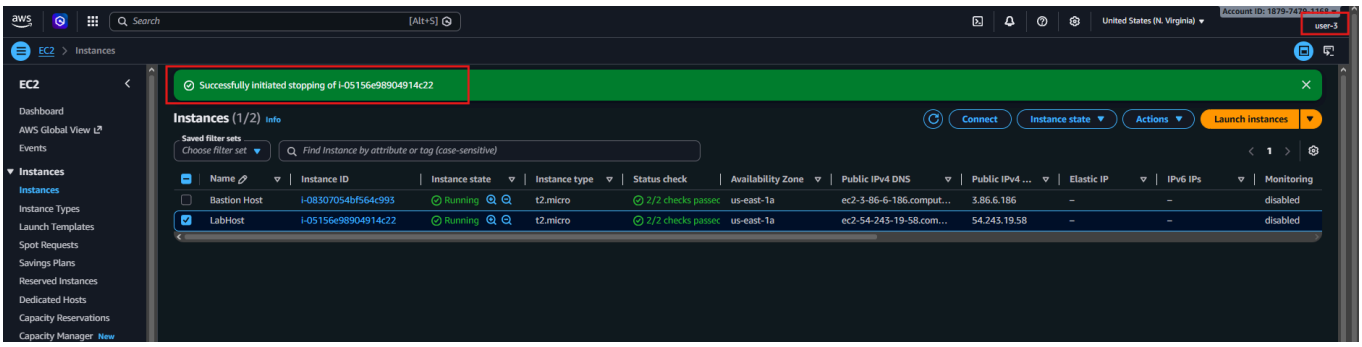


Figure 06. The instance will successfully enter the Stopping state like below

This confirms that user-3 can perform the instance-management actions allowed by the attached policy.

Close the private browser window.

Lab 2: Build your VPC and Launch a Web Server

Task 1: Create Your VPC

In the AWS search bar, search for and open VPC.

Verify that the selected Region is N. Virginia (us-east-1).

From the VPC console, select VPC Dashboard, then choose Create VPC.

If Create VPC is not available, select Launch VPC Wizard instead.

Configure the VPC using the following settings:

Resources to create: VPC and more

Name tag auto-generation: Auto-generate = lab

IPv4 CIDR block: 10.0.0.0/16

Availability Zones: 1

Public subnets: 1

Private subnets: 1

Expand Customize subnets CIDR blocks and configure:

Public subnet CIDR: 10.0.0.0/24

Private subnet CIDR: 10.0.1.0/24

Configure the remaining settings:

NAT gateways: In 1 AZ

VPC endpoints: None

DNS hostnames: Enabled

DNS resolution: Enabled

Review the preview panel and confirm the following resources will be created:

VPC: lab-vpc

Public subnet: lab-subnet-public1-us-east-1a

Private subnet: lab-subnet-private1-us-east-1a

Route tables: lab-rtb-public and lab-rtb-private1-us-east-1a

Internet Gateway: lab-igw

NAT Gateway: lab-nat-public1-us-east-1a

Select Create VPC.

Wait for all resources to finish provisioning. The NAT Gateway may take several minutes to become available.

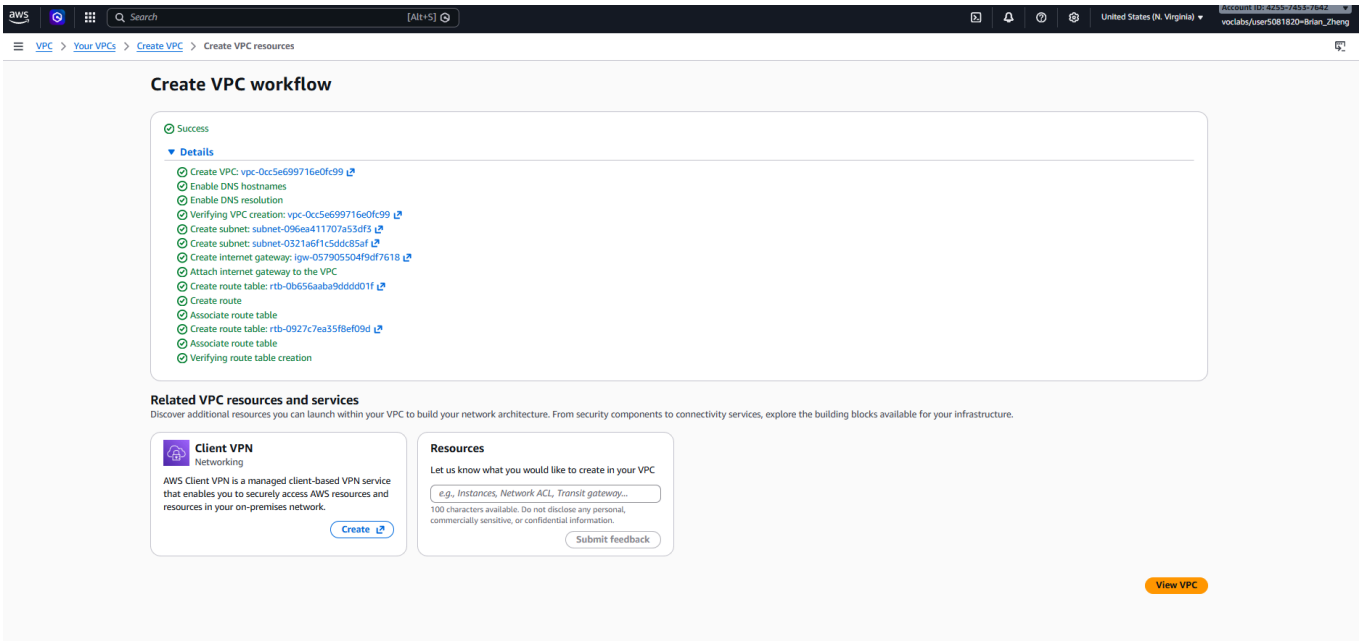


Figure 07. Wait for all resources to finish provisioning. The NAT Gateway may take several minutes to become available.

Once deployment is complete, select View VPC.

Review the resources created by the wizard:

One VPC

One public subnet

One private subnet

Separate route tables for each subnet

One Internet Gateway

One NAT Gateway

Explore the VPC resources using the navigation pane. For example:

Select Subnets to view subnet details.

Select Route tables to review routing configurations.

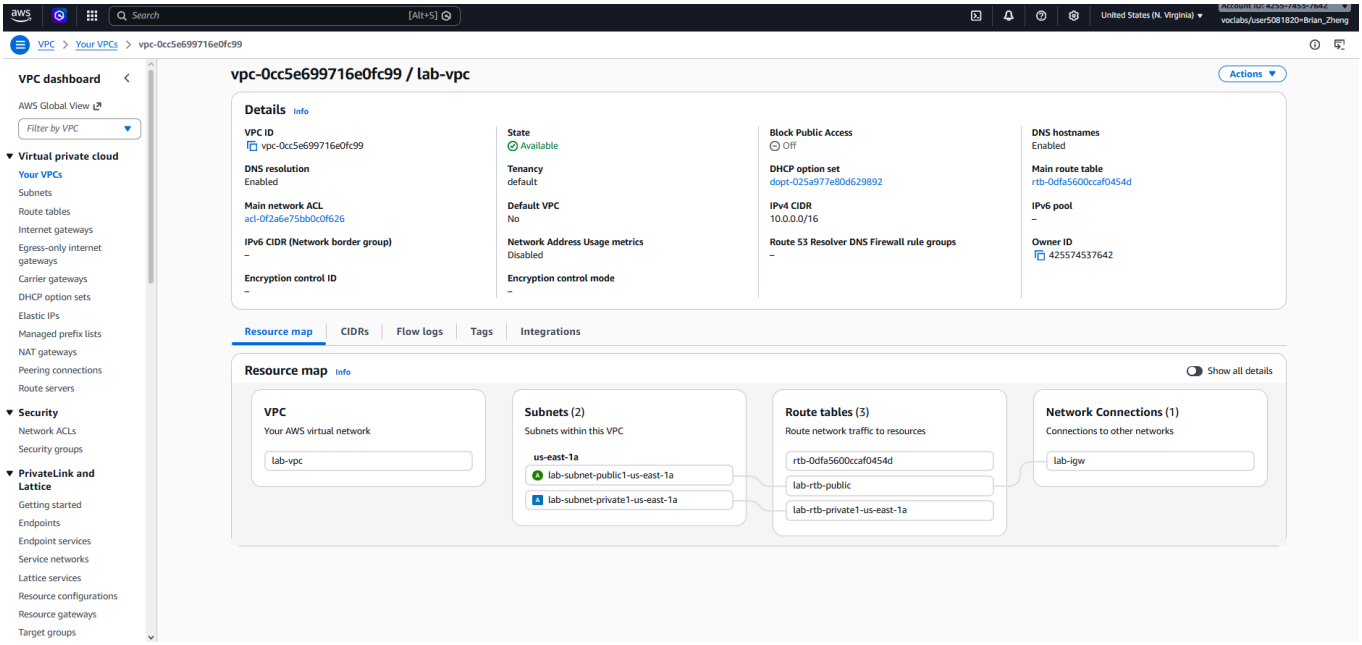


Figure 08. Select Route tables to review routing configurations.

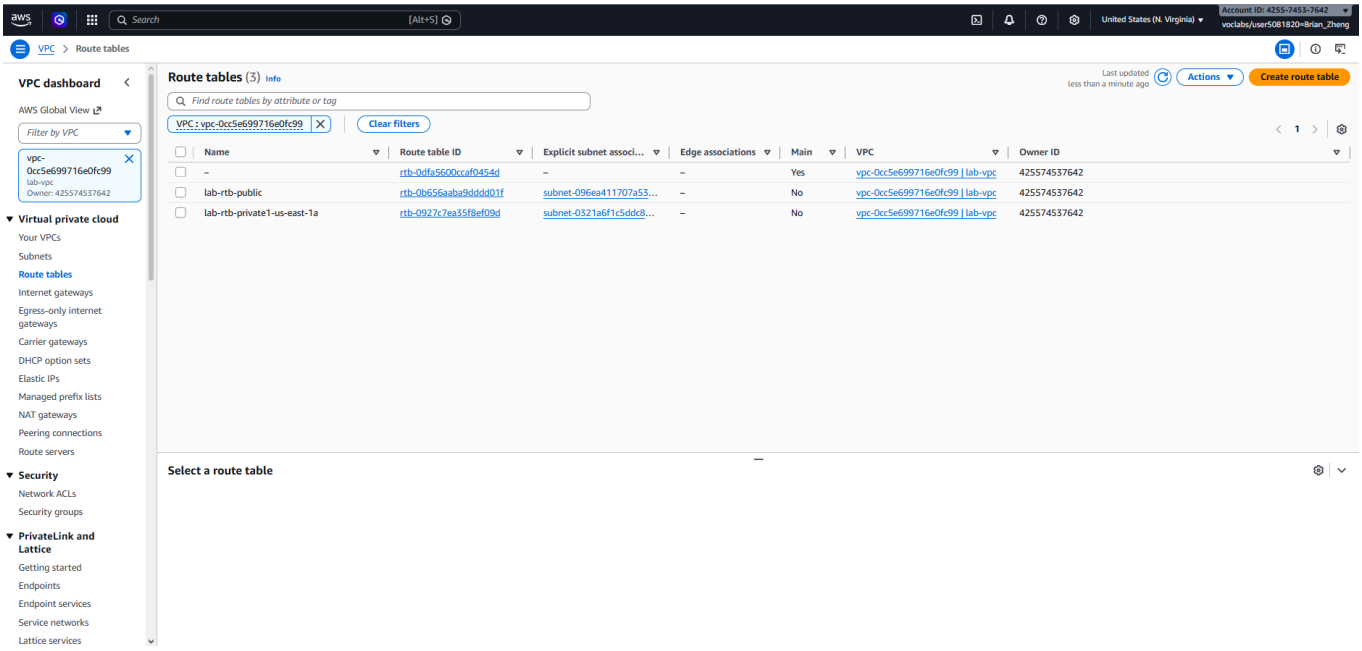


Figure 09. Select Route tables to review routing configurations.

Understanding the Network Configuration:

The Internet Gateway enables communication between resources in the VPC and the public internet.

The public subnet (10.0.0.0/24) contains addresses in the range 10.0.0.x.

It is considered public because its route table sends internet-bound traffic (0.0.0.0/0) to the Internet Gateway.

The NAT Gateway allows instances in private subnets to access the internet without being directly reachable from the internet.

The private subnet (10.0.1.0/24) contains addresses in the range 10.0.1.x and uses the NAT Gateway for outbound internet access.

Task 2: Create Additional Subnets

In the left navigation pane, select Subnets.

Create a Second Public Subnet:

Choose Create subnet and configure:

VPC ID: lab-vpc

Subnet name: lab-subnet-public2

Availability Zone: Select the second Availability Zone (for example, us-east-1b)

IPv4 CIDR block: 10.0.2.0/24

Select Create subnet.

This subnet will use IP addresses in the range 10.0.2.x.

Create a Second Private Subnet:

Choose Create subnet again and configure:

VPC ID: lab-vpc

Subnet name: lab-subnet-private2

Availability Zone: Select the second Availability Zone (for example, us-east-1b)

IPv4 CIDR block: 10.0.3.0/24

Select Create subnet.

This subnet will use IP addresses in the range 10.0.3.x.

It should look similar to this after these steps:

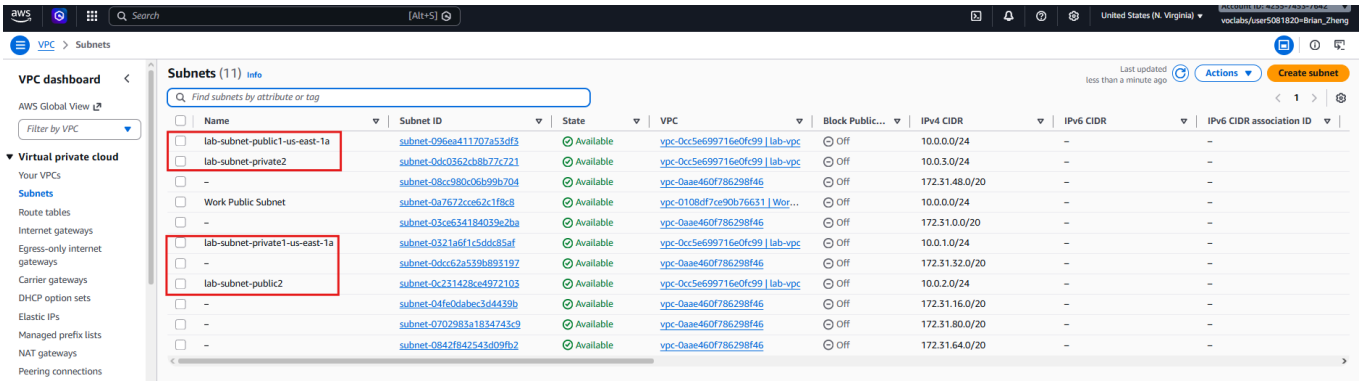


Figure 10. It should look similar to this after these steps

Associate the Private Route Table:

In the left navigation pane, select Route tables.

Select the lab-rtb-private1-us-east-1a route table.

If it is not visible, refresh the page.

Open the Routes tab.

Notice that the destination 0.0.0.0/0 points to a NAT Gateway.

This allows instances in private subnets to access the internet without being publicly accessible.

Open the Subnet associations tab.

Select Edit subnet associations.

Keep lab-subnet-private1-us-east-1a selected and also select lab-subnet-private2.

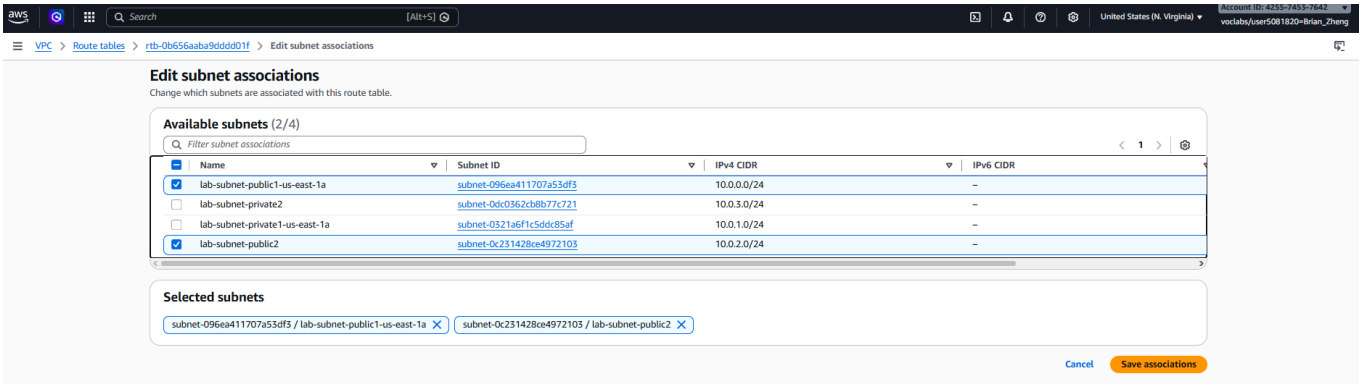


Figure 11. Keep lab-subnet-private1-us-east-1a selected and also select lab-subnet-private2.

Choose Save associations.

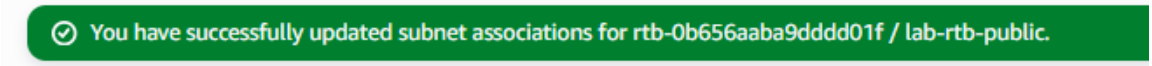


Figure 12. Choose Save associations.

Both private subnets now use the same route table and NAT Gateway.

Associate the Public Route Table:

Select the lab-rtb-public route table.

Open the Routes tab.

Notice that the destination 0.0.0.0/0 points to an Internet Gateway (IGW).

This allows resources in public subnets to communicate directly with the internet.

Open the Subnet associations tab.

Select Edit subnet associations.

Keep lab-subnet-public1-us-east-1a selected and also select lab-subnet-public2.

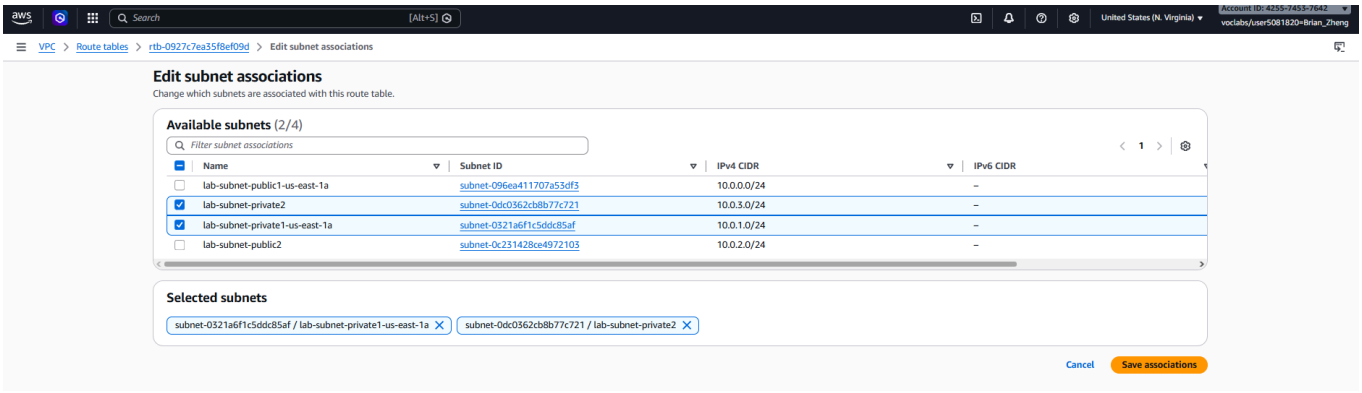


Figure 13. Keep lab-subnet-public1-us-east-1a selected and also select lab-subnet-public2.

Choose Save associations.



Figure 14. Choose Save associations.

Result:

Your VPC now contains:

Two public subnets across two Availability Zones

Two private subnets across two Availability Zones

A public route table connected to the Internet Gateway

A private route table connected to the NAT Gateway

The new subnets are now configured to route network traffic correctly based on whether they are public or private.

Task 3: Create a VPC Security Group

In the left navigation pane, select Security groups.

Choose Create security group and configure:

Security group name: Web Security Group

Description: Enable HTTP access

VPC: lab-vpc

Under Inbound rules, select Add rule and configure:

Type: HTTP

Source: Anywhere-IPv4

Description: Permit web requests

After these two steps it should look like this:

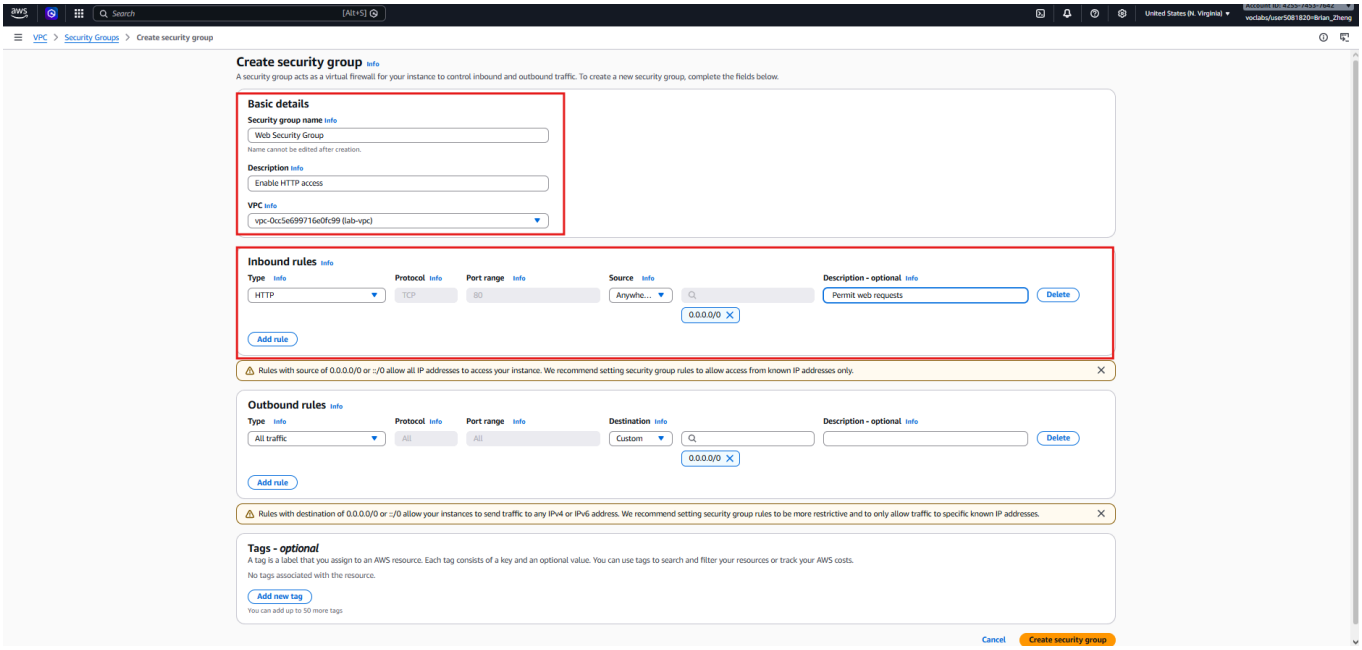


Figure 15. After these two steps it should look like this

Scroll to the bottom of the page and choose Create security group.

The Web Security Group will be used when launching the EC2 web server.

Task 4: Launch a Web Server Instance

Search for and open EC2 from the AWS search bar.

Select Launch instance.

Configure the instance:

Name: Web Server 1

AWS automatically creates a Name tag using this value.

Under Application and OS Images (AMI):

Keep the default Amazon Linux 2023 AMI selected.

Under Instance type:

Keep the default t2.micro instance type.

Under Key pair:

Select vockey.

Under Network settings, choose Edit and configure:

Network: lab-vpc

Subnet: lab-subnet-public2

Auto-assign public IP: Enable

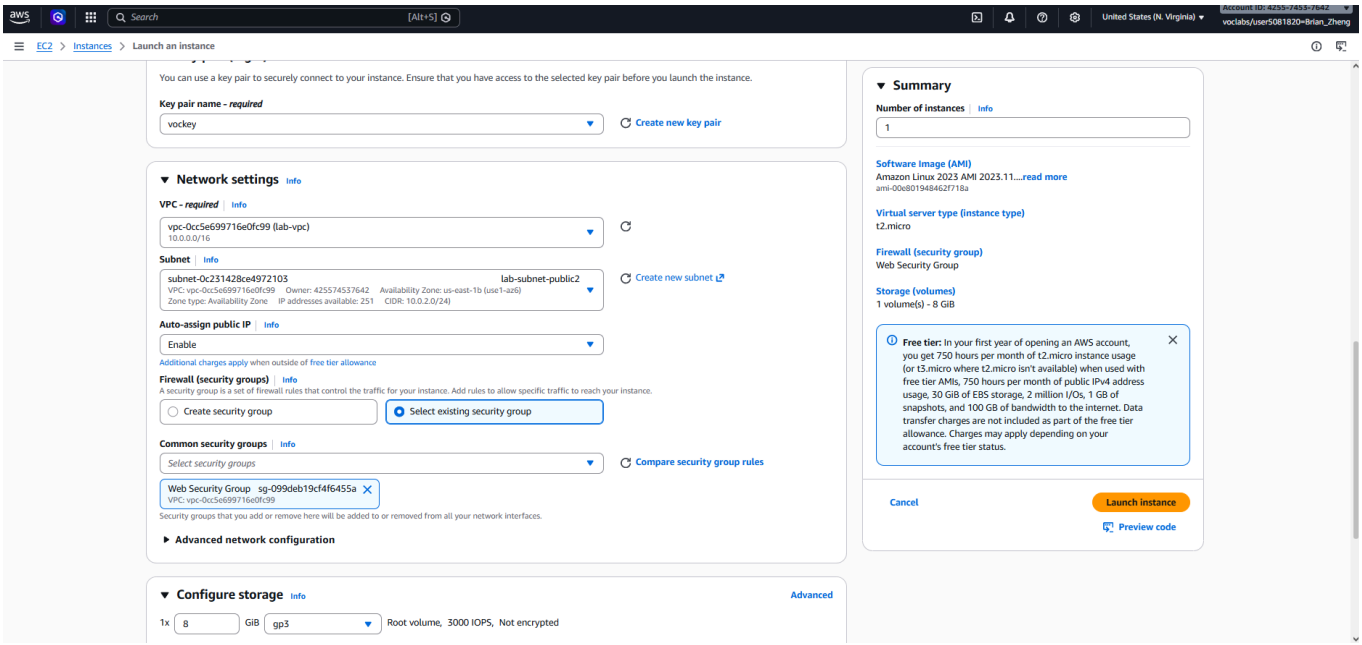


Figure 16. Auto-assign public IP: Enable

Under Firewall (security groups):

Select Choose existing security group

Select Web Security Group

Leave the default storage settings unchanged.

Configure User Data

Expand Advanced details and scroll to User data.

Paste the following script into the User data box:

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

This script automatically:

Installs Apache, PHP, and database components

Downloads the lab application

Configures and starts the web server

Review the settings and choose Launch instance.

When the success message appears, select View all instances.

Verify the Web Server

Wait until Web Server 1 displays 2/2 checks passed in the Status Check column.

Refresh the page periodically to monitor progress.

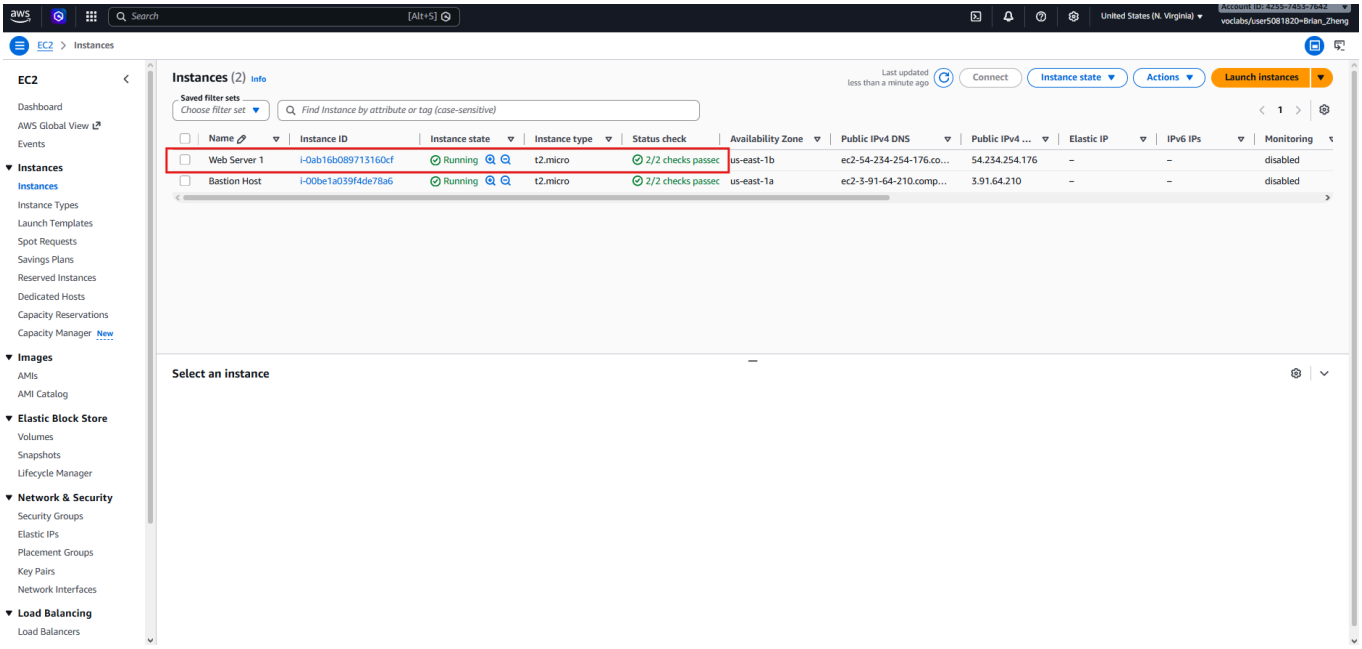


Figure 17. Refresh the page periodically to monitor progress.

Select Web Server 1.

In the Details tab, copy the Public IPv4 DNS value.

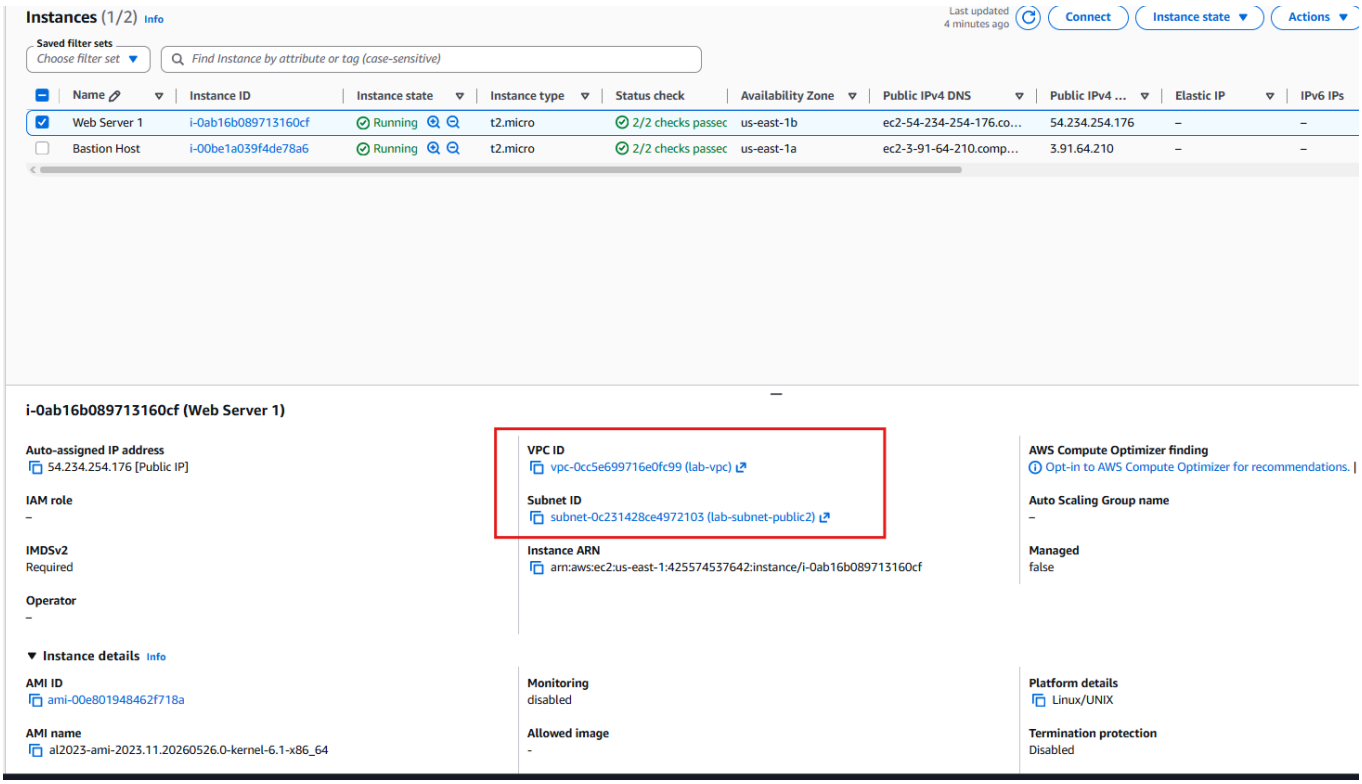


Figure 18. In the Details tab, copy the Public IPv4 DNS value.

Open a new browser tab, paste the DNS address, and press Enter.

Verify that the web page loads successfully and displays the AWS logo along with instance metadata.

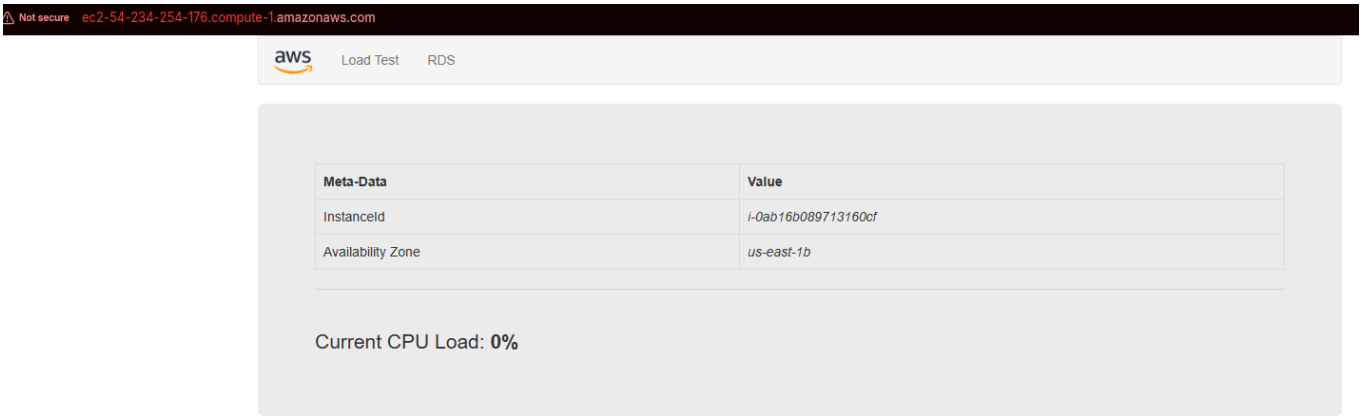


Figure 19. Verify that the web page loads successfully and displays the AWS logo along with instance metadata.

The deployed architecture now includes:

A custom VPC

Public and private subnets across two Availability Zones

Internet and NAT Gateways

Security Groups

An EC2 instance hosting a web application

Lab 3: Introduction to Amazon EC2

Task 1: Launch Your Amazon EC2 Instance

Launch an Amazon EC2 Instance

Open EC2 from the AWS Management Console.

Verify that the selected Region is N. Virginia (us-east-1).

Select Launch instance.

Configure the Instance

Under Name and tags, enter:

Name: Web Server

AWS will automatically create a Name tag for the instance using this value.

Under Application and OS Images (AMI):

Keep the default Amazon Linux 2023 AMI selected.

The AMI determines the operating system and software configuration used by the instance.

Under Instance type:

Keep the default t2.micro instance type selected.

This instance type includes 1 vCPU and 1 GiB of memory.

Under Key pair (login):

Select vockey.

This key pair enables secure SSH access to the instance if needed.

Configure Networking

Under Network settings, choose Edit.

Configure:

VPC: Lab VPC

Subnet: PublicSubnet1

Auto-assign public IP: Enabled

Under Firewall (security groups):

Select Create security group

Configure:

Security group name: Web Server security group

Description: Security group for my web server

Under Inbound security group rules, remove the default rule.

Configure Storage

Leave the default storage settings unchanged.

The instance will use an 8 GiB Amazon EBS root volume.

Configure Advanced Settings

Expand Advanced details.

Under Termination protection, select Enable.

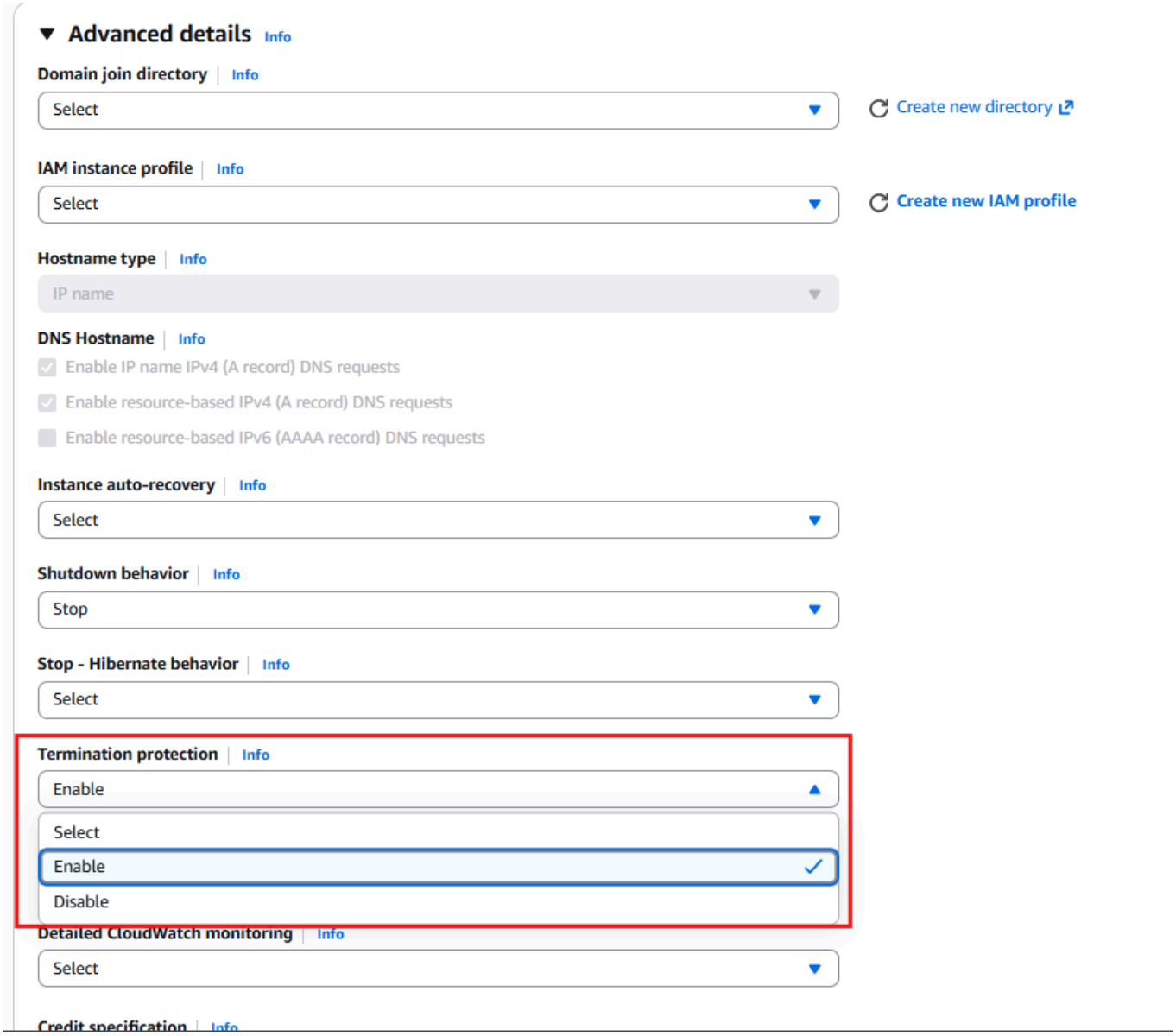


Figure 20. Under Termination protection, select Enable.

This prevents the instance from being accidentally terminated.

Scroll to the User data section and paste the following script:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

This script will:

Install Apache HTTP Server

Configure the service to start automatically

Start the web server

Create a simple web page

Launch and Verify

Review the configuration and select Launch instance.

When the success message appears, choose View all instances.

Select the Web Server instance from the Instances list.

Review the information in the Details tab, including:

Instance type

Security group settings

Network configuration

Public IPv4 DNS address

Wait for the instance to reach the following status:

Instance State: Running

Status Checks: 2/2 checks passed

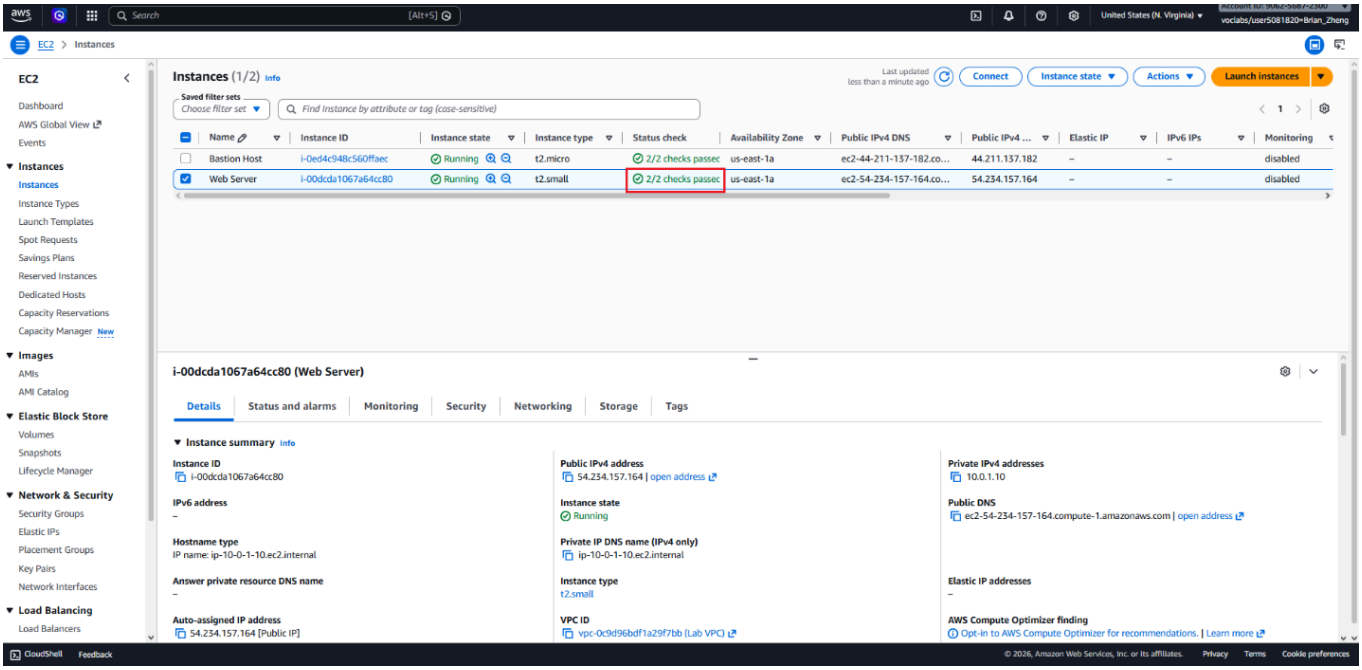


Figure 21. Status Checks: 2/2 checks passed

Once these checks are complete, the EC2 instance is fully operational and ready to host the web page created by the user data script.

Task 2: Monitor Your Instance

Monitor and Troubleshoot the Instance

Select the Status checks tab.

Verify that both checks show Passed:

System reachability

Instance reachability

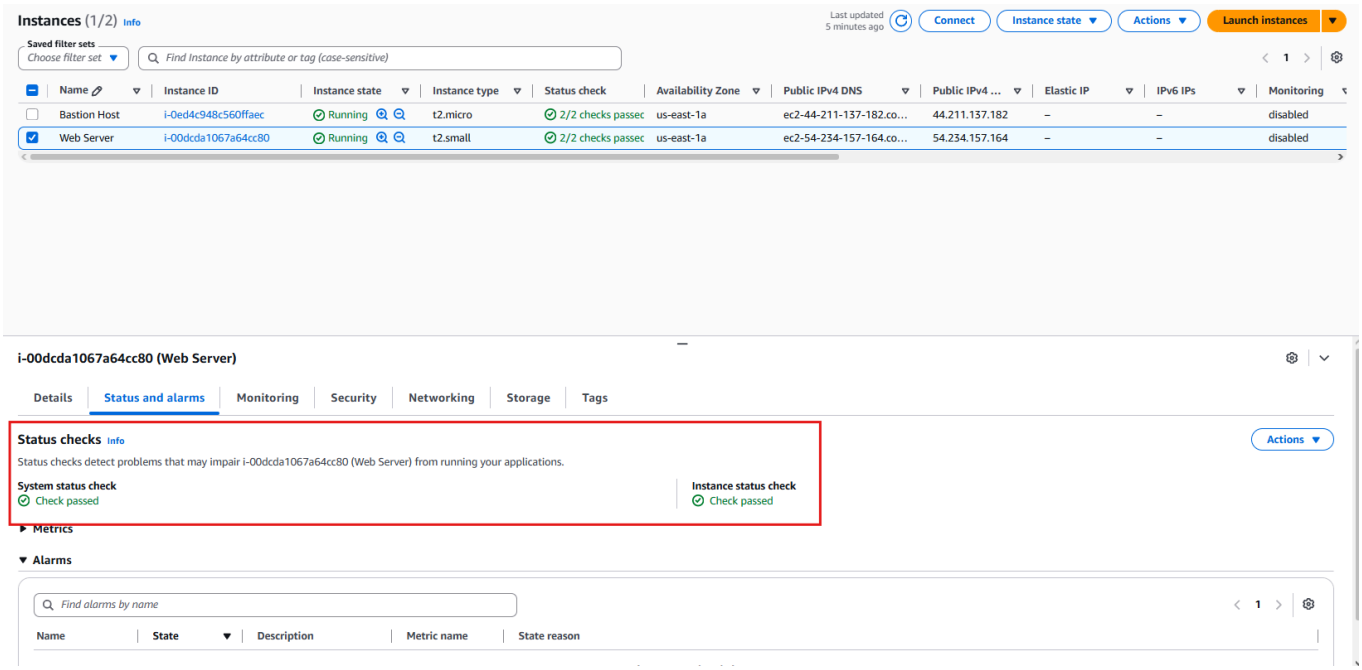


Figure 22. Instance reachability

These automated checks help identify hardware or software issues that could affect instance availability.

View CloudWatch Metrics

Open the Monitoring tab.

Review the Amazon CloudWatch metrics for the instance.

Since the instance was recently launched, only a limited amount of data may be available.

To view a larger version of a graph, select the three-dot menu and choose Enlarge.

By default, EC2 sends basic monitoring data to CloudWatch every five minutes. Detailed monitoring can be enabled for one-minute intervals.

View the System Log

From the Actions menu, select:

Monitor and troubleshoot

Get system log

Review the system log output.

The system log records console messages generated during startup.

It is useful for diagnosing boot, kernel, and service-related issues.

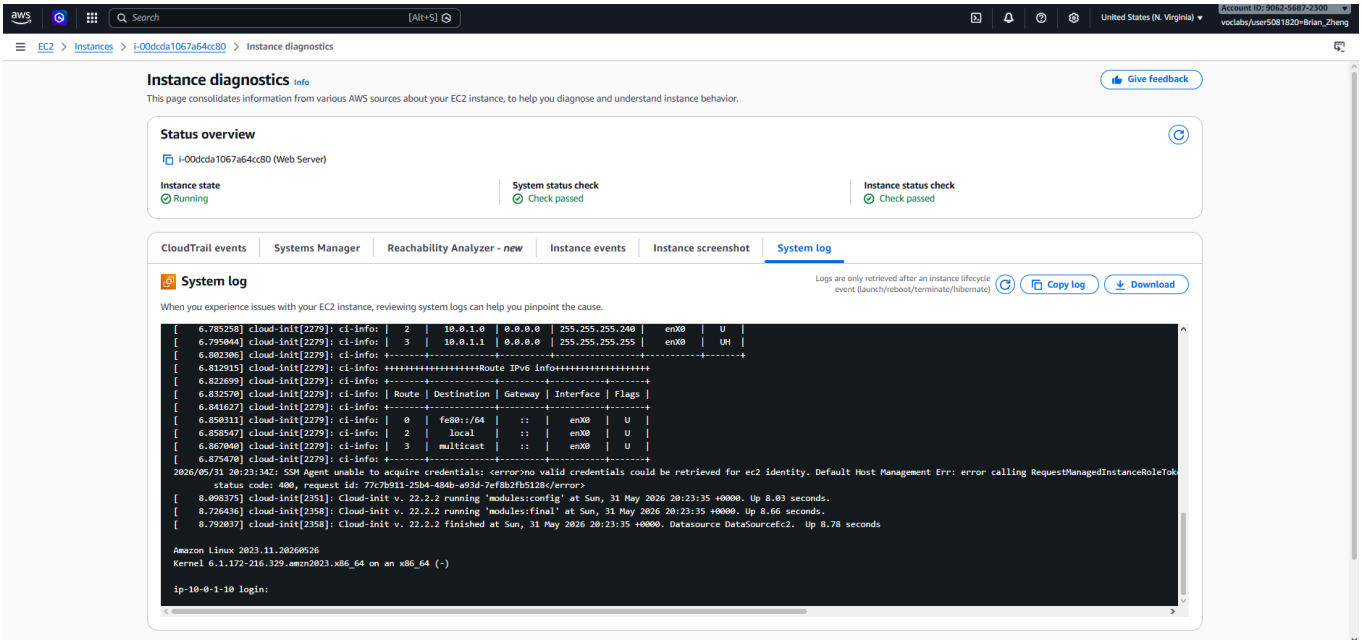


Figure 23. It is useful for diagnosing boot, kernel, and service-related issues.

Scroll through the log and verify that the Apache HTTP package was installed by the user data script.

Choose Cancel to close the log window.

View an Instance Screenshot

Ensure the Web Server instance is still selected.

From the Actions menu, select:

Monitor and troubleshoot

Get instance screenshot

Review the screenshot of the instance console.

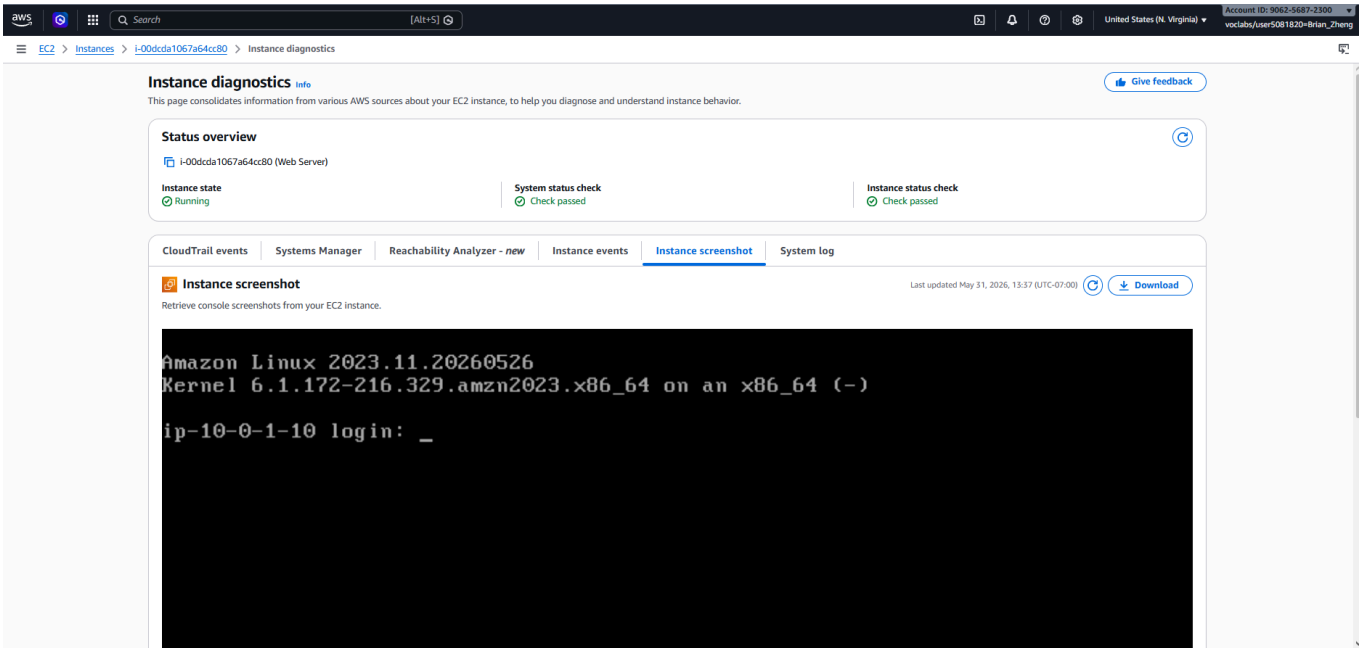


Figure 24. Review the screenshot of the instance console.

This displays what would appear on a monitor connected directly to the instance.

Instance screenshots can be useful when troubleshooting systems that are unreachable through SSH or Remote Desktop.

Choose Cancel to close the screenshot window.

Task 3: Update Your Security Group and Access the Web Server

Test Access to the Web Server

Ensure the Web Server instance is selected and open the Details tab.

Copy the instance's Public IPv4 address.

Open a new browser tab, paste the IP address into the address bar, and press Enter.

Notice that the web page does not load.

The security group currently does not allow inbound HTTP traffic on port 80 which is why web results are blocked.

Update the Security Group

Keep the browser tab open and return to the EC2 Console.

In the left navigation pane, select Security Groups.

Choose Web Server security group.

Open the Inbound rules tab.

Notice that there are currently no inbound rules configured.

Select Edit inbound rules, then choose Add rule.

Configure the rule:

Type: HTTP

Source: Anywhere-IPv4

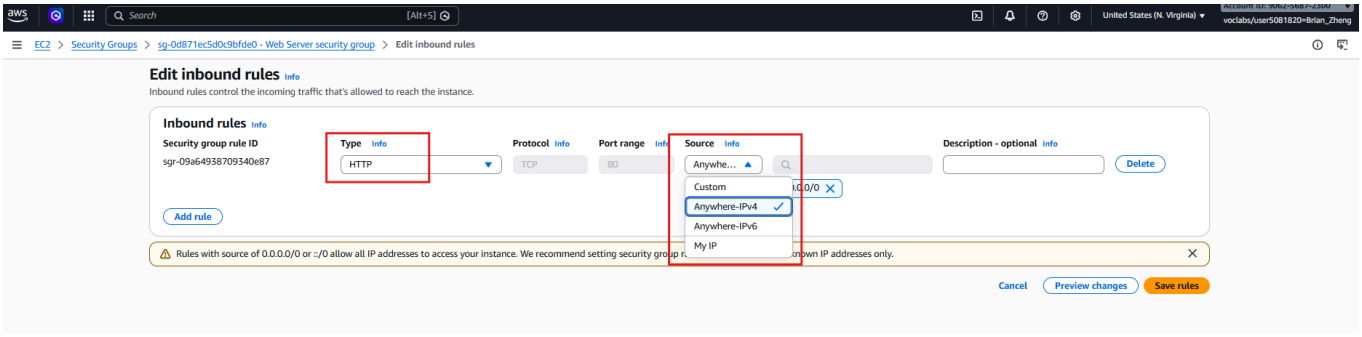


Figure 25. Source: Anywhere-IPv4

Choose Save rules.

Verify Access

Return to the browser tab containing the web server's IP address.

Refresh the page.

The web page should now load successfully and display:

Hello From Your Web Server!



Figure 26. Hello From Your Web Server!

Task 4: Resize Your Instance: Instance Type and EBS Volume

Stop the Instance

In the EC2 Console, select Instances and choose the Web Server instance.

From the Instance state menu, select Stop instance, then choose Stop to confirm.

Wait until the instance state changes to Stopped.

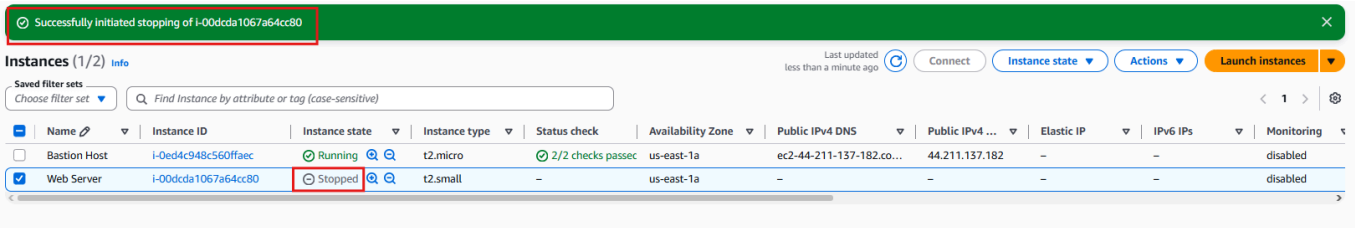


Figure 27. Wait until the instance state changes to Stopped.

A stopped instance does not incur compute charges.

Any attached EBS volumes continue to incur storage charges.

Resize the Instance

With the Web Server instance selected, open:

Actions → Instance settings → Change instance type

Set:

Instance type: t2.small

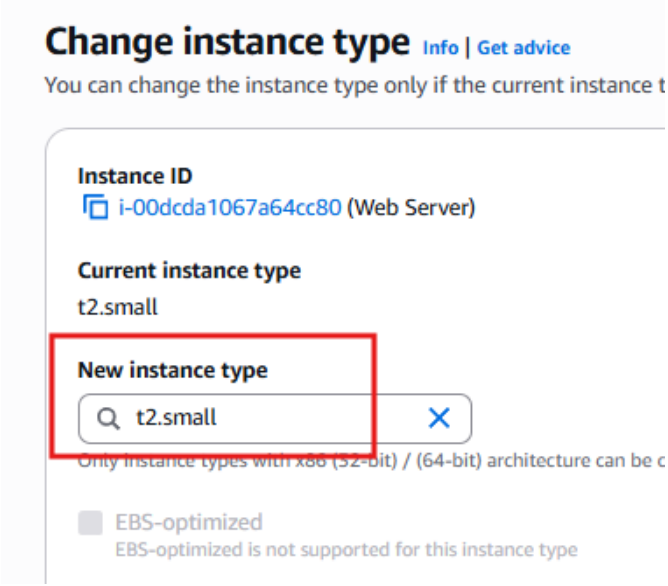


Figure 28. Instance type: t2.small

Choose Apply.

The t2.small instance provides twice the memory of a t2.micro instance.

Enable Stop Protection

Select:

Actions → Instance settings → Change stop protection

Enable Stop protection and save the change.

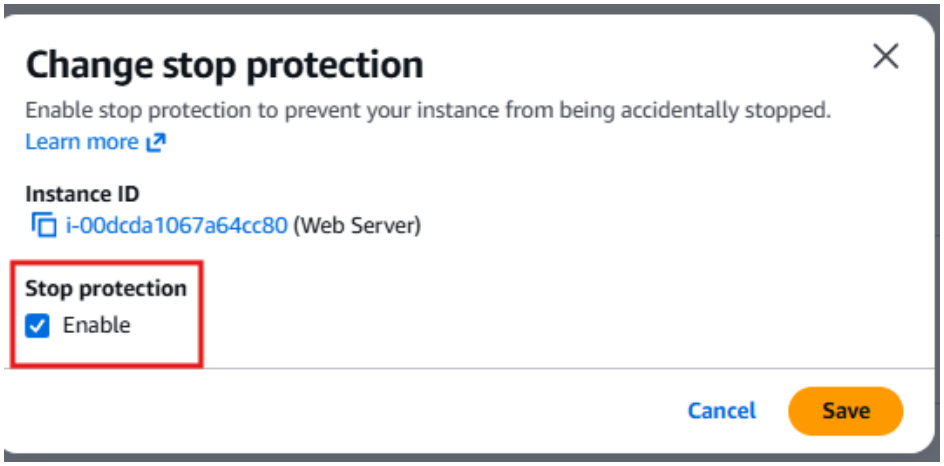


Figure 29. Enable Stop protection and save the change.

Stop protection helps prevent accidental shutdowns.

When an instance is stopped and restarted, it may receive a new public IPv4 address, but it retains its private IPv4 address and EBS volumes.

Increase the EBS Volume Size

Open the Storage tab and select the attached Volume ID.

Select the volume, then choose:

Actions → Modify volume

Change the volume size from 8 GiB to 10 GiB.

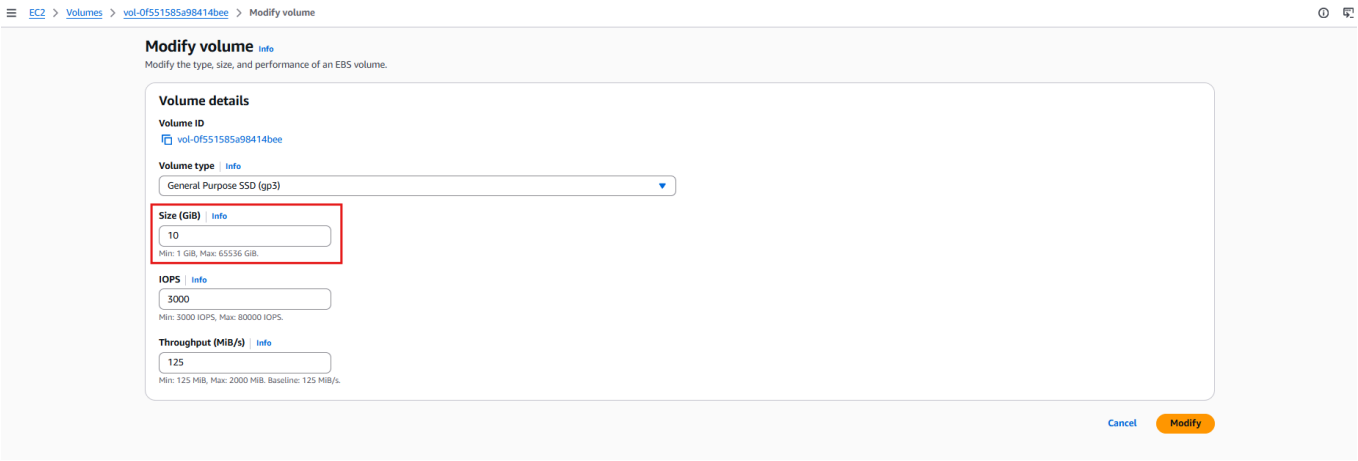


Figure 30. Change the volume size from 8 GiB to 10 GiB.

Choose Modify, then confirm by selecting Modify again.

Start the Updated Instance

Return to Instances.

Select the Web Server instance.

From the Instance state menu, choose Start instance.

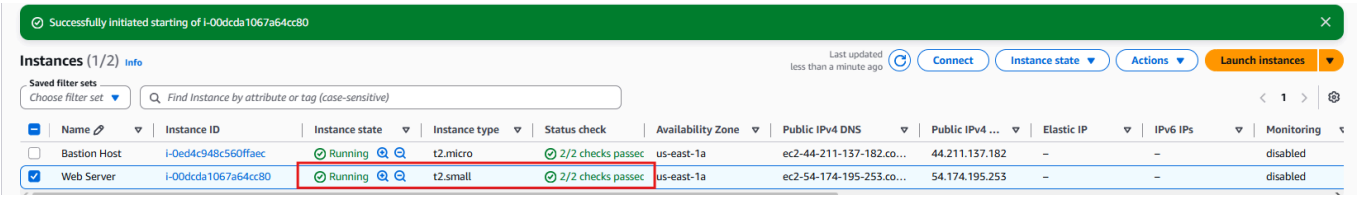


Figure 31. From the Instance state menu, choose Start instance.

The instance now runs with:

t2.small instance type

10 GiB EBS root volume

Task 5: Explore EC2 Limits

Search for and open Service Quotas.

In the navigation pane, select AWS services.

Search for and select Amazon Elastic Compute Cloud (Amazon EC2).

In the Find quotas search box, enter:

running on-demand:

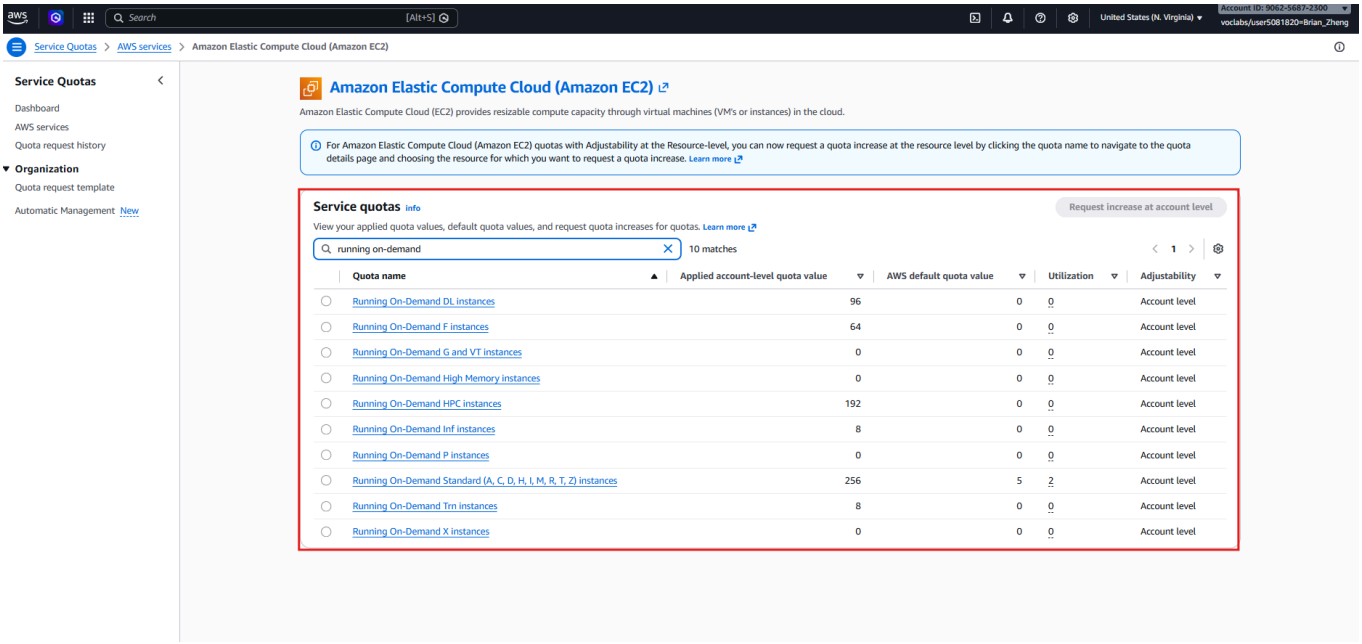


Figure 32. running on-demand

Review the displayed quotas.

These limits control the number and types of EC2 instances that can run within a region.

AWS account owners can request increases for many service quotas if needed.

Task 6: Test Stop Protection

Return to the EC2 Console and open Instances.

Select the Web Server instance.

From the Instance state menu, choose Stop instance, then select Stop.

Notice the error message indicating that the instance cannot be stopped.s

Notice the error message indicating that the instance cannot be stopped.

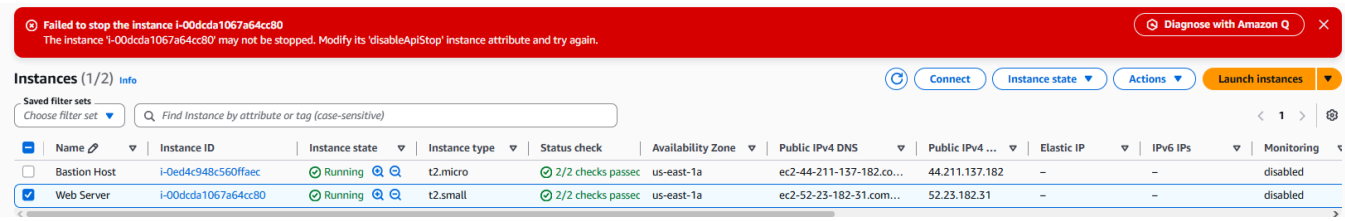


Figure 33. Notice the error message indicating that the instance cannot be stopped.

This occurs because Stop Protection is enabled.

Disable Stop Protection

Select:

Actions → Instance settings → Change stop protection

Clear the Enable checkbox.

Choose Save.

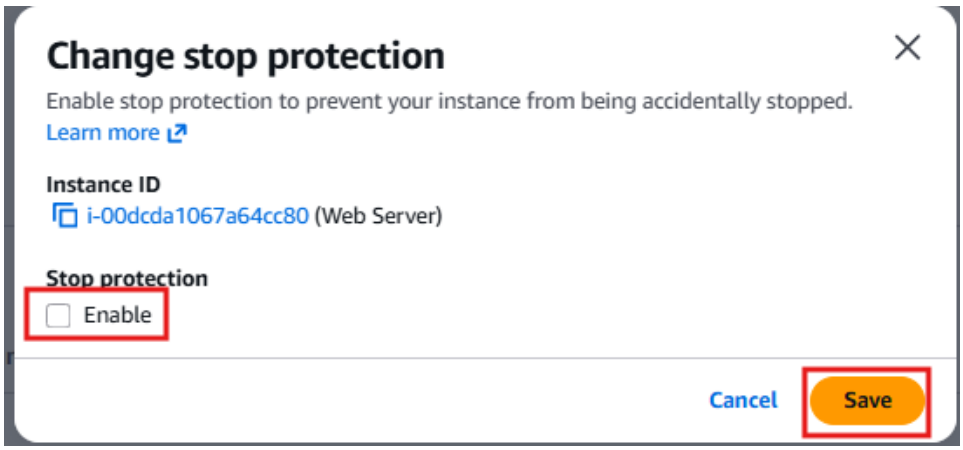


Figure 34. Choose Save.

Stop the Instance

Select the Web Server instance again.

From the Instance state menu, choose Stop instance.

Select Stop to confirm.

The instance will now stop successfully because stop protection has been disabled.

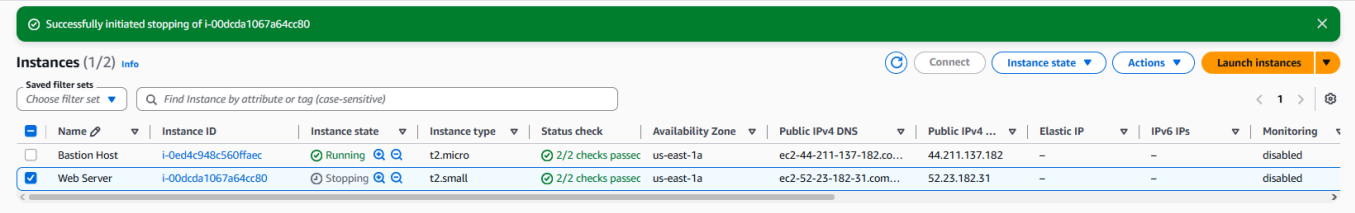


Figure 35. The instance will now stop successfully because stop protection has been disabled.

Troubleshooting and Lessons Learned

The AWS Academy grading interface did not consistently register completed steps. Rechecking the grading status during the exercise helped identify steps that needed to be repeated.

Conclusion

The exercises connected IAM permissions, VPC routing, and EC2 operations in a practical workflow. They reinforced permission testing, public and private subnet design, web-server deployment, instance monitoring, and lifecycle controls.